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SUPERSEDING
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May 31, 1966

FEDERAL SPECIFICATION

PIPE, PLASTIC (POLYETHYLENE, PE, SDR-PR)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers polyethylene (PE) pipe made from 5 plastic pipe materials in 5 standard design ratios (SDR) and 5 water pressure ratings (PR).

1.2 SDR ratio, PR rating and PE material. SDR, PR and PE material in table I shall be as specified (see 6.2, 6.4 and 6.5).

TABLE I. Pressure ratings

SDR	PE Pipe Materials									
	PE3006		PE3406		PE2306		PE2305		PE1404	
	p.s.i.	MN/m^2	p.s.i.	MN/m^2	p.s.i.	MN/m^2	p.s.i.	MN/m^2	p.s.i.	MN/m^2
5.3	200	1.38	200	1.38	200	1.38	160	1.10	125	0.86
7	160	1.10	160	1.10	160	1.10	125	0.86	100	.69
9	125	0.86	125	0.86	125	0.86	100	0.69	80	.55
11.5	100	.69	100	.69	100	.69	80	0.55	—	
15	80	.55	80	.55	80	.55	—		—	

1.2.1 Sizes. PE pipe shall be of the following nominal pipe sizes, as specified (see 6.2):

1/2, 3/4, 1, 1-1/4, 1-1/2, 2, 2-1/2, 3, 4 and 6 inches

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-B-601 - Boxes; Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.
PPP-T-60 - Tape; Packaging, Waterproof.
PPP-T-97 - Tape; Pressure-Sensitive Adhesive, Filament Reinforced.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

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(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standard:

D 2239 - Specification for Polyethylene (PE) Plastic Pipe (SDR-PR).

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Sanitation Foundation Testing Laboratory, Inc. (NSF) Standard:

No. 14 - Thermoplastic Materials, Pipe, Fittings, Valves, Joining Materials.

(Application for copies should be addressed to the National Sanitation Foundation Testing Laboratory, Inc., P.O. Box 1468, Ann Arbor, MI 48106.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Union Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 General. PE pipe shall conform to the requirements of ASTM D 2239, and shall be for potable water, or other water and fluids, as specified (see 4.3.1 and 6.2).

3.2 Pipe for potable water, public health and safety hazards.

3.2.1 Each contractor shall submit to the contracting agency proof that the PE pipe he proposes to supply under this specification conforms to the requirements of the National Sanitation Foundation Standard No. 14. The label, or listing with reexamination, of the National Sanitation Foundation may be accepted as evidence that the PE pipe conforms to this requirement.

3.2.2 In lieu of the label, or listing with reexamination, the contractor may submit independent proof, satisfactory to the contracting agency, that the PE pipe conforms to the applicable requirements of the published standards, including methods of test, of the National Sanitation Foundation Standard No. 14.

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3.2.3 Compliance with the above preliminary requirements in regard to public health and safety hazards does not absolve the bidder from complete compliance with the requirements of this specification in order to secure acceptance of his material.

3.3 Pipe for other water, fluids or other use. PE pipe for other than potable water service may include, in addition to clean rework material, clean rejected or damaged pipe that has not been in service.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling and inspection. Inspection levels and acceptable quality levels (AQL's) shall be in accordance with MIL-STD-105.

4.2.1 Lot. Unless otherwise specified (see 6.2), a lot shall consist of each SDR, PR and PE material submitted for inspection at the same time.

4.2.2 Dimensional and visual examination of end item. PE pipe shall be examined for these characteristics in accordance with level S-3 and an AQL of 2.5 percent defective for major defects and 4.0 percent defective for minor defects listed in table II.

TABLE II. Classification of defects, end item

Defects	Major	Minor
SDR, PR, and PE material not as specified	X	
Cracks or holes	X	
Foreign inclusions	X	
Eccentricity more than 12 percent	X	
Dimensions not within tolerance	X	
Identification marking omitted, incorrect, illegible		X
Damage or defects affecting function or serviceability	X	
Damage or defects not affecting function or serviceability		X

4.2.3 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the packaging, packing, and marking comply with the requirements in section 5. Defects shall be scored in accordance with table III. For examination of interior packaging, the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 with an AQL of 4.0 defects per hundred units.

TABLE III. Classification of preparation for delivery defects

Examine	Defects
Marking (exterior and interior)	Omitted; incorrect; illegible, improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Distortion of container.

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4.2.4 Dimensional measurements. Measure diameters, wall thickness and eccentricity in accordance with ASTM D2230.

4.3 Test methods.

4.3.1 General. Test pipe in accordance with ASTM D2239 (see 3.1).

4.3.2 Sustained pressure and environmental cracking tests. Due to the length of time required for conducting of these tests, they shall be conducted only when the NSF label, or listing with re-examination, has been withheld from the PE pipe submitted for acceptance by the Government. In the absence of the NSF label, or listing with reexamination, the tests shall be conducted in accordance with ASTM D2239.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Plastic pipe, with a nominal size of 2 inches or less, shall be coiled to a minimum safe diameter and the coils secured with not less than 5 evenly spaced ties of soft wire, cotton tape, or with 4 ties conforming to PPP-T-97, type III. Prior to coiling, the ends of the coils shall be sealed with plastic plugs or with tape conforming to PPP-T-60, type III, class 1. Noncoiled pipe shall have ends sealed by means that will withstand the hazards of transportation, without damage and will prohibit the entrance of foreign matter to the interior of the pipe.

5.1.2 Level C. The pipe shall be packaged in a manner to provide necessary protection against damage from the supply source to the first receiving activity, except that when pipe is specified for use with potable-water (see 6.2), the ends shall be sealed as specified in 5.1.1.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Pipe of like description shall be packed in boxes conforming to PPP-B-601, overseas type, PPP-B-621, class 2; or PPP-B-636, class weather-resistant. The boxes shall be closed and strapped in accordance with the appendix to the applicable box specification. The gross weight of the PPP-B-636 box shall not exceed the weight limitations of the box specification of wood boxes 200 pounds. When a single coil of tubing exceeds 250 pounds, each coil packed in a wood box, shall be modified by the installation of skids in accordance with PPP-B-621.

5.2.2 Level B. Pipe of like description shall be packed as specified in 5.2.1, except the containers shall be the domestic class or type and unless otherwise specified (see 6.2) strapping shall not be required.

5.2.3 Level C. The plastic pipe shall be packed to insure carrier acceptance and safe delivery to destination at lowest rates in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking.

5.3.1 Civil agencies. In addition to marking required by the contract or order, the interior packaging and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to markings required by the contract or order, the interior packaging and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. PE pipe is primarily intended for conveying potable water. It is also used for sprinkler systems, pump installations, air-conditioning water lines, and irrigation and drainage systems.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) PE material, SDR and PR (see 1.2).
(c) Pipe size (see 1.2.2).
(d) Whether pipe shall be for potable water or other use (see 3.1 and 5.1.2).
(e) Size of lot, if different from 4.2.1.
(f) Selection of applicable level of packaging and packing required (see 5.1 and 5.2).
(g) Whether strapping is required (see 5.2.2).

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6.3 For civil agency procurements. Whe level B packaging is required by civil agency procurements, the requirements in 5.1.1, level A packaging shall apply.

6.4 Type designation. The type designation code for PE pipe materials is illustrated in the example below:

PE 2305:

PE	23	0	5
Type of plastic	Material properties of P23 in ASTM D1248	Hydrostatic design stress less than 1000 p.s.i.	Hydrostatic design stress in hundreds to one significant figure

6.5 Relation between standard dimension ratio, hydrostatic design stress, and pressure rating. The following equations, commonly known as the ISO equations, relate standard dimension ratio, hydrostatic design stress, and pressure rating:

$$2S/P = R + 1 \quad \text{or} \quad 2S/P = Di/t = 1$$

where:

S = hydrostatic design stress in MN/m^2 or p.s.i.
P = pressure rating in MN/m^2 or p.s.i.
Di = average inside diameter in mm or inches.
t = minimum wall thickness in mm or inch.
R = standard thermoplastic pipe dimension ratio Di/t, also known as SDR.

Preparing Activity:

GSA-FSS

Orders for this publication are to be placed with the General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 10 cents each.

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